

University of Al-Kafeel

College of Dentistry



LEC-2

Autonomic Nervous System Pharmacology (ANS)

Asst.Prof.Dr.Yasmeen Ali Hussein

By the end of this lecture, you should be able to



- **Describe the organization and divisions of the Autonomic Nervous System .**
- **Identify the major neurotransmitters and receptors involved in ANS signaling.**
- **Explain the mechanisms of action and effects of autonomic drugs.**
- **Correlate autonomic pharmacology with clinical and dental practice.**

INTRODUCTION TO ANS

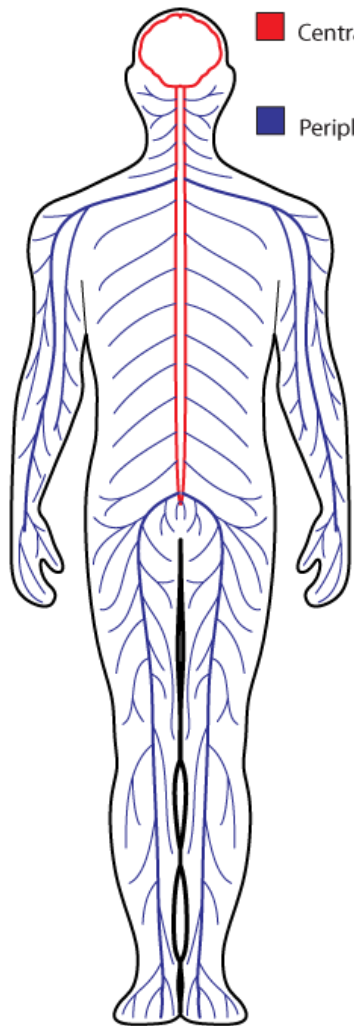
- (NS) can be defined as the system of cells, tissues, and organs that regulates the body's responses to internal and external stimuli.
- The nervous system is divided into the *central nervous system* (CNS) and *peripheral nervous system*
- The peripheral nervous system is divided into *somatic (voluntary)* and *the autonomic system –ANS-(involuntary)*
- ANS has two divisions: **SNS** and **PNS**



INTRODUCTION TO ANS

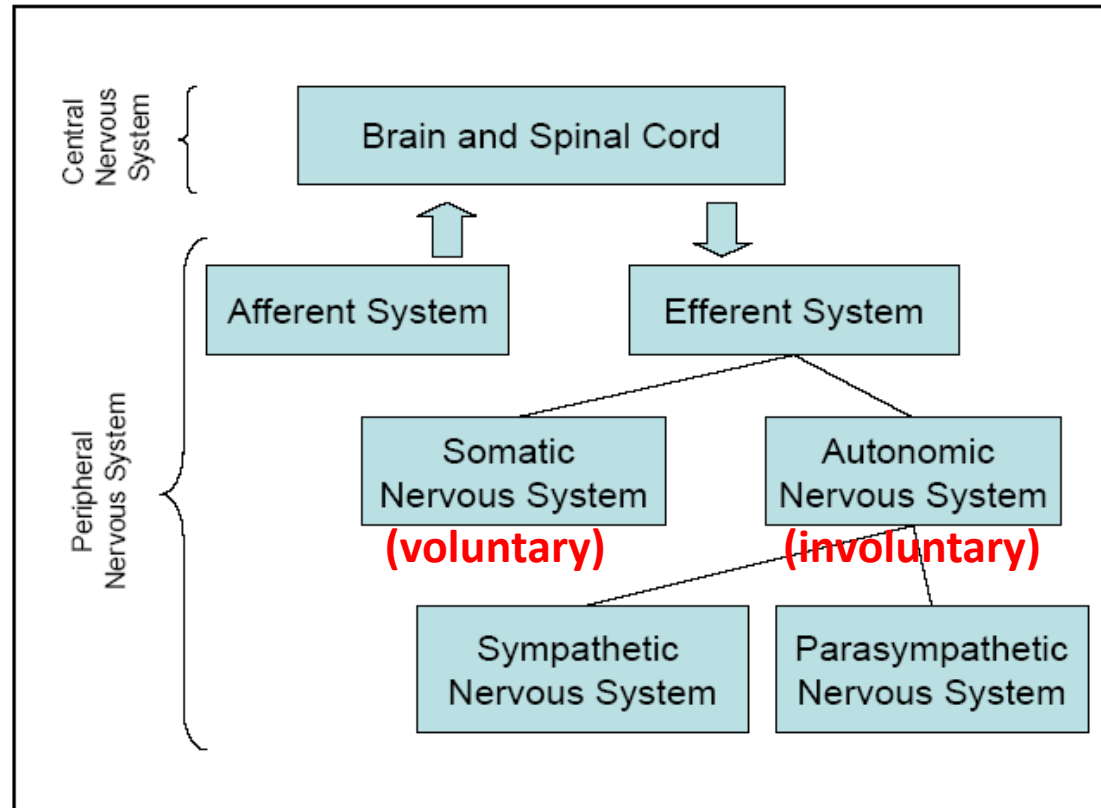
- ANS is a network consisting of nerves, ganglia, and plexuses that maintains body homeostasis by integrating signals from somatic and visceral sensors to modulate tissue and organ functions
- ANS provides the innervations to smooth muscle, cardiac muscle, and various glands





■ Central Nervous System

■ Peripheral Nervous System



INTRODUCTION TO ANS

❑ The autonomic nervous system (ANS)

- ANS conveys all the outputs from the central nervous system to the rest of the body, except for the motor innervation of skeletal muscle
- Skeletal muscle is innervated by somatic nerves, controlling voluntary actions
- It is very important to understand drugs that effect **autonomic nervous system.**

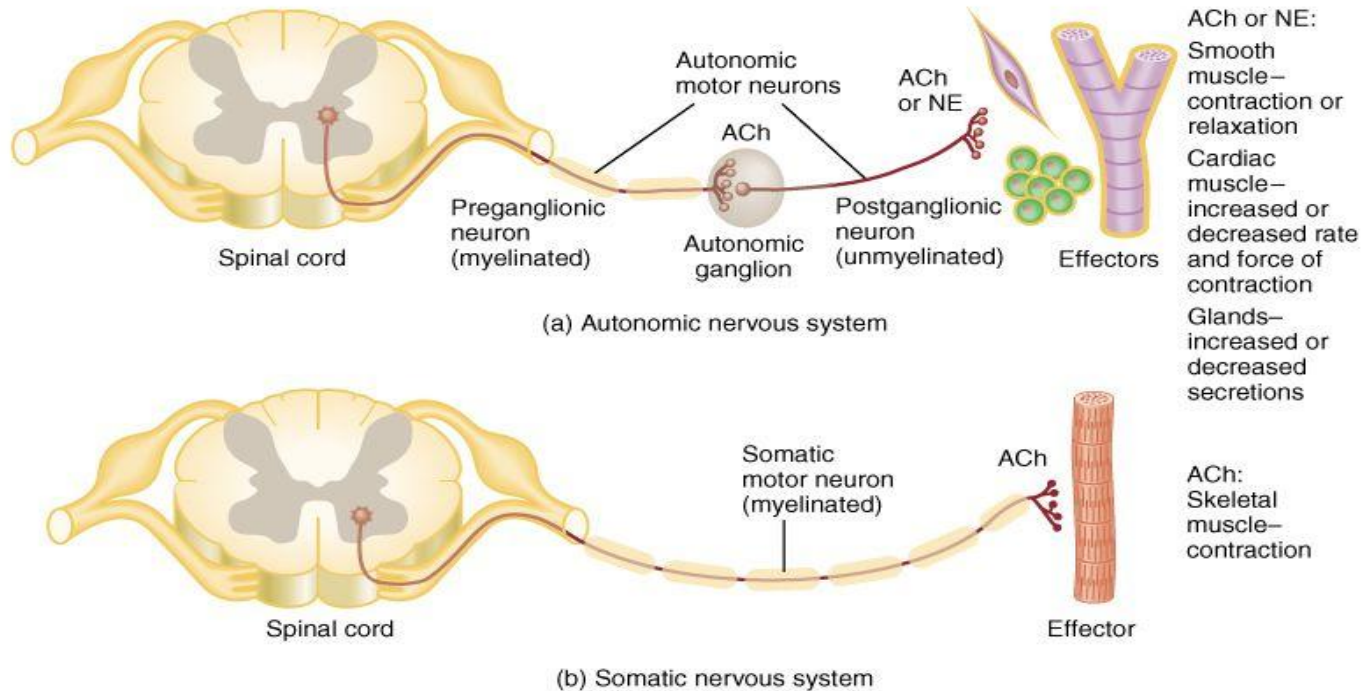


Differences between Somatic and Autonomic Innervation

Feature	Autonomic Nervous System (ANS)	Somatic Nervous System (SNS)
Number of neurons from spinal cord to effector	Two neurons: preganglionic and postganglionic	One neuron only
Presence of ganglion	Present (autonomic ganglia between neurons)	Absent
Myelination	Preganglionic fiber – myelinated; Postganglionic fiber – unmyelinated	Entire motor neuron is myelinated
Neurotransmitter released	Acetylcholine (ACh) or Norepinephrine (NE)	Acetylcholine (ACh) only
Receptor type at effector	Muscarinic or Adrenergic receptors	Nicotinic (Nm) receptors
Effector organs	Smooth muscle, cardiac muscle, glands	Skeletal (voluntary) muscle
Type of control	Involuntary (unconscious control)	Voluntary (conscious control)
Main function	Regulates internal body functions (heart rate, digestion, secretion)	Controls body movements (movement, posture)



Structural and Functional Differences Between Autonomic and Somatic Nervous System



► Anatomical Aspects of ANS

There is 2 major subdivision:-

- **Parasympathetic ANS (PANS)**
- **Sympathetic ANS (SANS)**
- **Enteric Nervous System (ENS)** is the semiautonomous part of ANS located in the GIT responsible for absorption and secretion in GIT



1.Parasympathetic nervous system

Dominate roles under normal physiological condition

- Rest and Digest**



Parasympathetic nervous system

It has an important function in maintaining the internal environment of the human body in a steady state. This role is vital in returning the body to a homeostatic state after trauma.

As various changes occur within the environment, both internal and external, the ANS reacts by regulating such things as the Blood Pressure, Heart Rate, Concentration of salts in the Blood Stream etc.

For example when the level of nutrients within the blood stream falls the Sensory Nervous System (SeNS) picks this up and stimulates the ANS to activate food seeking behaviour and also stimulates the digestive processes. If the body becomes dehydrated such as playing football on a very hot cloudless day the SeNS will pick up sensory information on the depletion of body fluid and the ANS will activate the mechanisms which conserve and replenish body fluids.

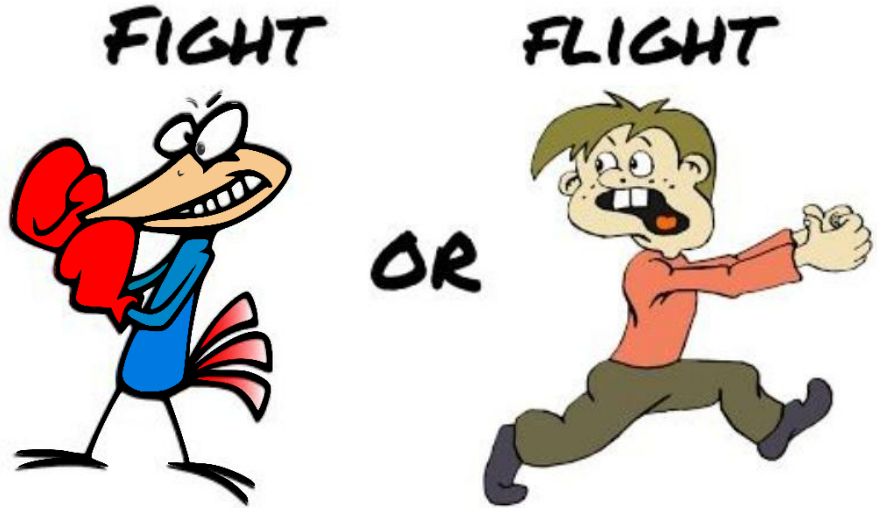
The ANS is also involved in many other body activities such as, waste disposal, response to stress, and sexual response.

The functions of the ANS underlie the physiological aspects of coping during stress and forms a major link between the nervous system and the endocrine system during these times.

2. Sympathetic nervous system

prepares organism to handle various environmental stresses

•Fight or Flight



2. Sympathetic nervous system

The autonomic nervous system (ANS) as we have already seen is part of the peripheral nervous system (PNS).

It has an important function in maintaining the internal environment of the human body in a steady state. This role is vital in returning the body to a homeostatic state after trauma.

As various changes occur within the environment, both internal and external, the ANS reacts by regulating such things as the Blood Pressure, Heart Rate, Concentration of salts in the Blood Stream etc.

For example when the level of nutrients within the blood stream falls the Sensory Nervous System (SeNS) picks this up and stimulates the ANS to activate food seeking behaviour and also stimulates the digestive processes. If the body becomes dehydrated such as playing football on a very hot cloudless day the SeNS will pick up sensory information on the depletion of body fluid and the ANS will activate the mechanisms which conserve and replenish body fluids.

The ANS is also involved in many other body activities such as, waste disposal, response to stress, and sexual response.

The functions of the ANS underlie the physiological aspects of coping during stress and forms a major link between the nervous system and the endocrine system during these times.

❑ Physiological antagonists

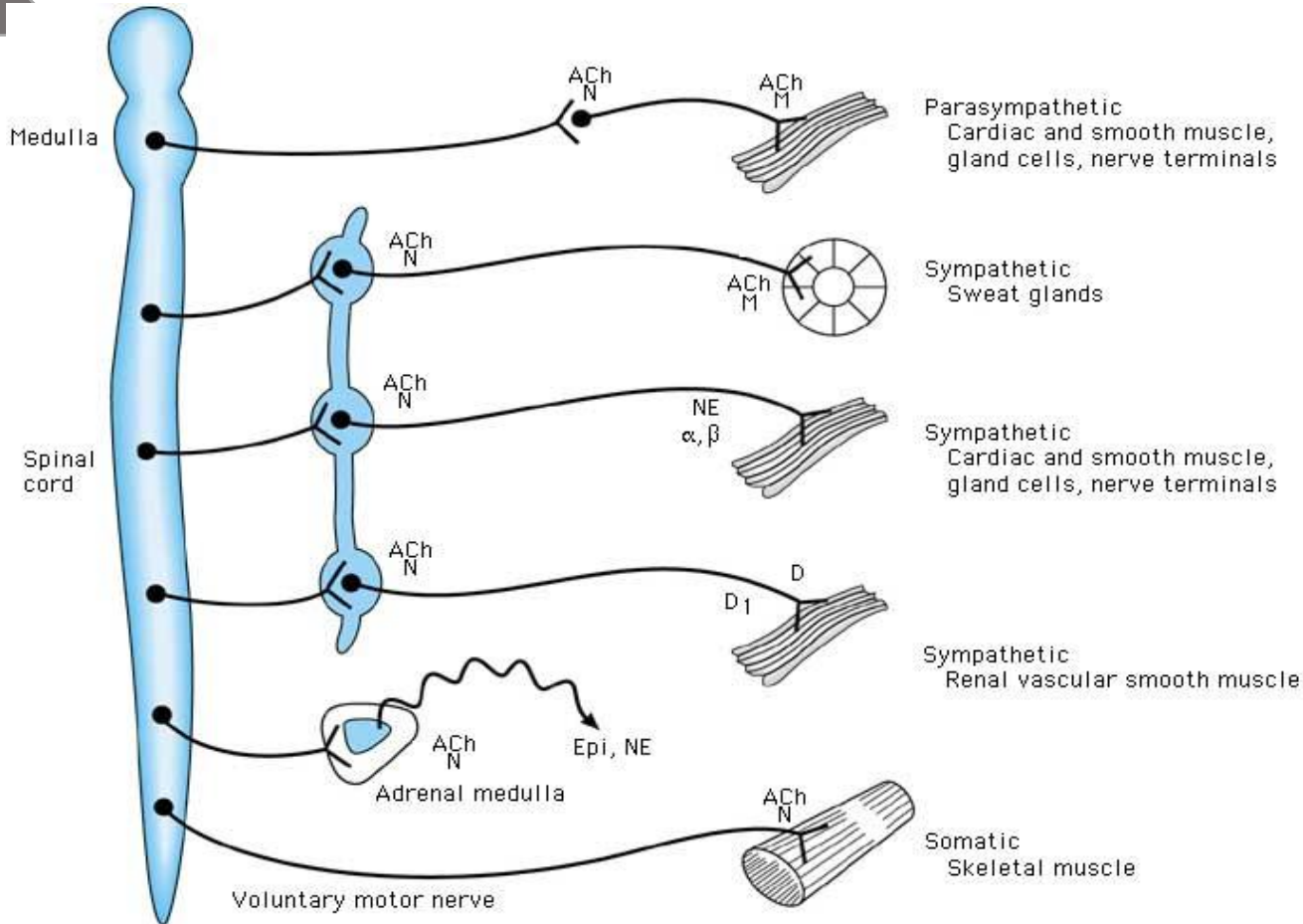
- The sympathetic and parasympathetic systems usually do not function independently; i.e., they are ***physiological antagonists***
- Often when one system inhibits a process, the other system will augment the level of activity so that the total response depends on the influence of both systems, **although this is not always the case (Spleen, Sweat Glands, Piloerector Muscles and Most Blood Vessels are controlled by SNS)**



- All parasympathetic post ganglionic fibres contain acetylcholine (Ach)
- *Most sympathetic postganglionic fibres contain noradrenaline (NA)*
The exception is postganglionic fibres innervating the skin, which use Ach
- All preganglionic fibres (both sympathetic and parasympathetic) contain Ach
- In the somatic nervous system, only a single fibre goes straight to the skeletal muscle and releases Ach.



ANS NEUROTRANSMISSION



Schematic diagram comparing some anatomic and neurotransmitter features of autonomic and somatic motor nerves.

Only the primary transmitter substances are shown.

Parasympathetic ganglia are found in or near the wall of the organ innervated.

Note that some sympathetic postganglionic fibers release acetylcholine or dopamine rather than norepinephrine.

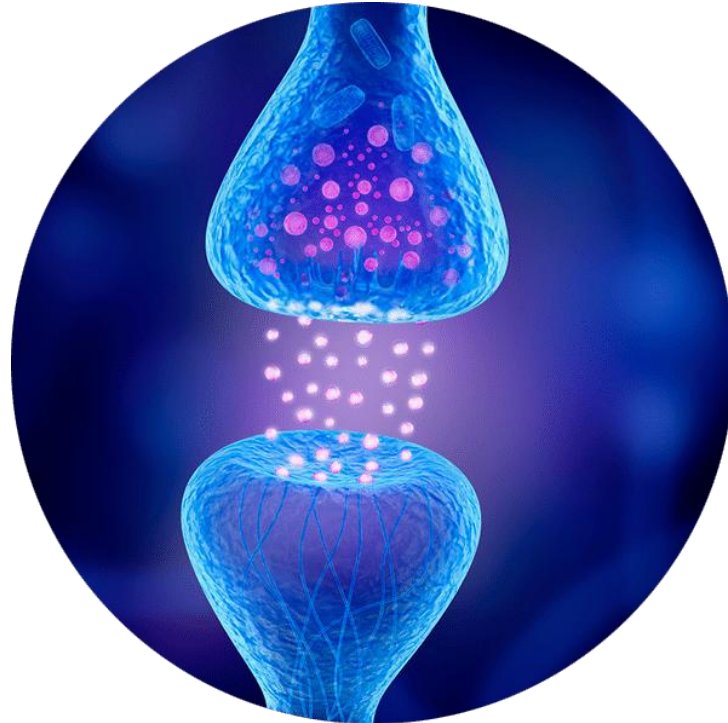
The adrenal medulla, a modified sympathetic ganglion, receives sympathetic preganglionic fibers and releases epinephrine and norepinephrine into the blood.

(ACh, acetylcholine; D, dopamine; Epi, epinephrine; NE, norepinephrine; N, nicotinic receptors; M, muscarinic receptors.)

ANS NEUROTRANSMISSION

System	Neurotransmitter at Ganglion	Neurotransmitter at Effector	Receptor at Effector	Main Effect
Parasympathetic	ACh	ACh	Muscarinic (M)	↓ heart rate, ↑ secretion, ↑ smooth muscle contraction
Sympathetic (typical)	ACh	NE	α, β	↑ heart rate, vasoconstriction, bronchodilation
Sympathetic (sweat glands)	ACh	ACh	Muscarinic (M)	↑ sweating
Sympathetic (renal)	ACh	Dopamine	D ₁	Vasodilation in kidney
Sympathetic (adrenal medulla)	ACh	— (releases Epi, NE into blood)	—	General sympathetic activation
Somatic	—	ACh	Nicotinic (Nm)	Skeletal muscle contraction

Receptors Characteristics



A-Cholinoceptors

1-Nicotinic Receptors

- Respond to ACh
- Respond to nicotine (another ACh mimic)
- Do not respond to muscarine
- Act through Ligand-gated ion channels
- Agonist effects blocked by **tubocurarine**
- Receptor activation results in:
 - Rapid increases of Na^+ and Ca^{2+} conductance
 - Depolarization
 - Excitation



2 Subtypes of nicotinic receptors:

➤ **Nn-neuronal (ANS ganglia)**

are in sympathetic and parasympathetic ganglia (effects blocked with ganglionic blockers (e.g., **trimethaphan**, **hexamethonium**))

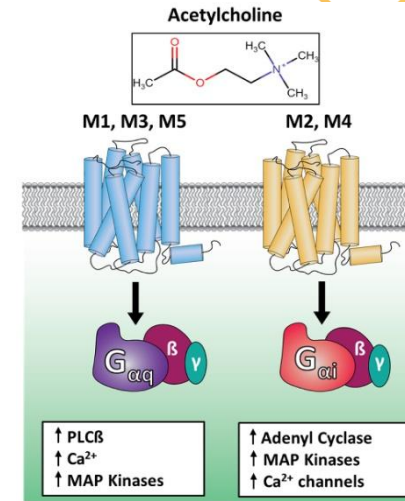
➤ **Nm-neuromuscular endplate (skeletal muscle)**

are at the neuromuscular junction (effects blocked with neuromuscular blockers (e.g., curare))



2-Muscarinic Receptors

- G-protein coupled receptor system
 - Slower responses
 - Agonist effects blocked by atropine
 - Respond to muscarine (an alkaloid)
 - Respond to ACh
 - Mimics the effects of parasympathetic
- At least **five** receptor subtypes have been described by molecular cloning.
(3 receptors are important M1 M2 M3)
 - **M1-(excitatory)** nerve endings
 - **M2-(inhibitory)** heart, some nerve endings
 - **M3-(excitatory)** effector cells, smooth muscle, exocrine glands, endothelium



CHOLINOCEPTORS

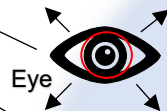
M₁	Nerve endings, Stomach parietal cells	G-coupled	Increase IP ₃
M₂	Heart, some nerve endings	G-coupled	Decrease cAMP, activates K ⁺ channel
M₃	Effector cells, smooth muscle, glands, endothelium	G-coupled	Increase IP ₃
N_n	ANS ganglia	Ion channel	Depolarizes, evokes action potential
N_m	Neuromuscular end plates	Ion channel	Depolarizes, evokes action potential



M_1 receptors

M_3 receptors

M_2 receptors



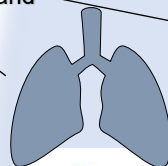
μ_3

Miosis



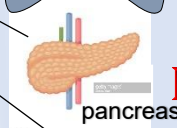
M_3

↑ Salivation



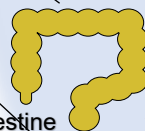
M_3

Bronchoconstriction



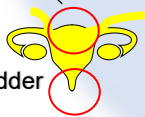
M_3

Secretion of the digestive enzymes



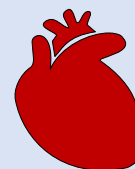
M_3

Constriction of smooth muscle



M_3

Urination



Heart

↓ contractility

↓ Heart rate

↓ Conductivity



M_1

Increase acid secretion



CHOLINOCEPTORS

PARASYMPATHETIC NERVES

"Rest and digest"

Constrict pupils

Stimulate saliva

Slow heartbeat

Constrict airways

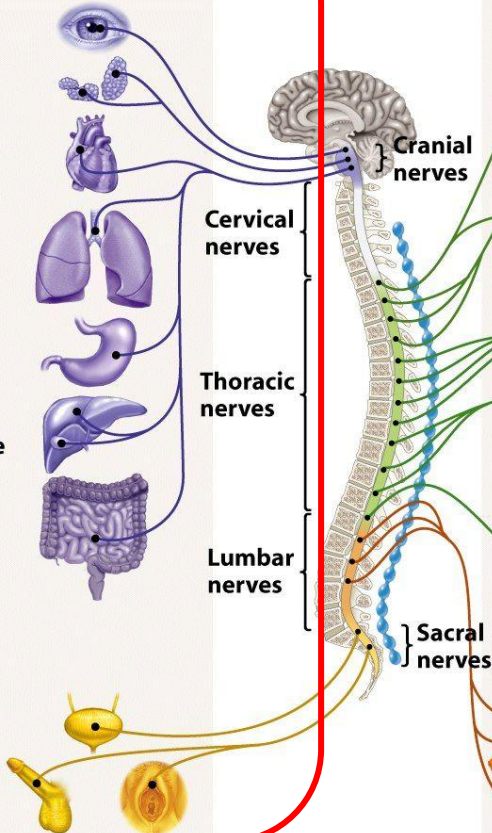
Stimulate activity of stomach

Inhibit release of glucose; stimulate gallbladder

Stimulate activity of intestines

Contract bladder

Promote erection of genitals



SYMPATHETIC NERVES

"Fight or flight"

Dilate pupils

Inhibit salivation

Increase heartbeat

Relax airways

Inhibit activity of stomach

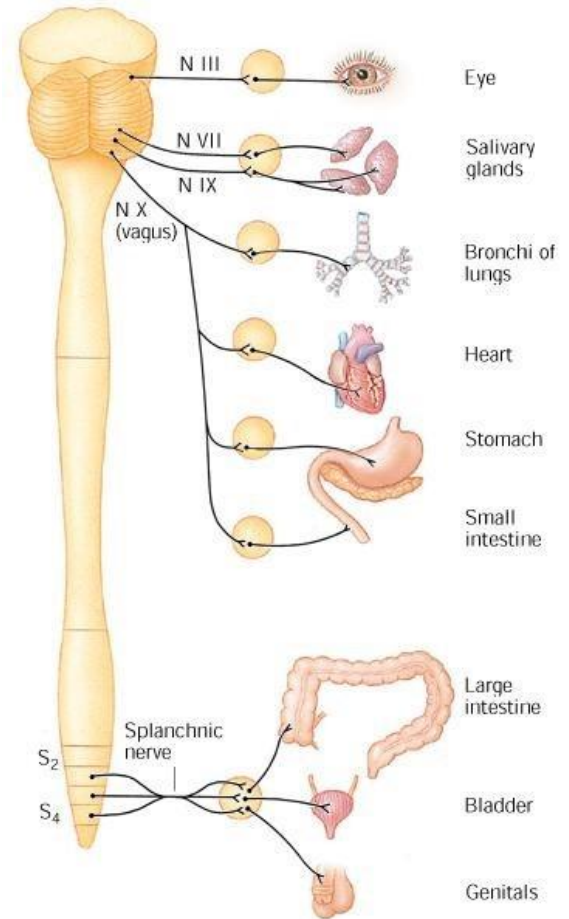
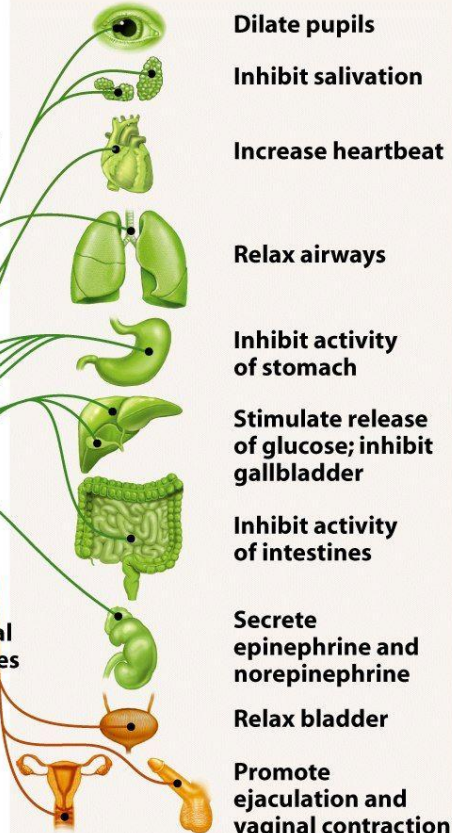
Stimulate release of glucose; inhibit gallbladder

Inhibit activity of intestines

Secrete epinephrine and norepinephrine

Relax bladder

Promote ejaculation and vaginal contraction



B-Adrenoceptors:

PARASYMPATHETIC NERVES

"Rest and digest"

Constrict pupils

Stimulate saliva

Slow heartbeat

Constrict airways

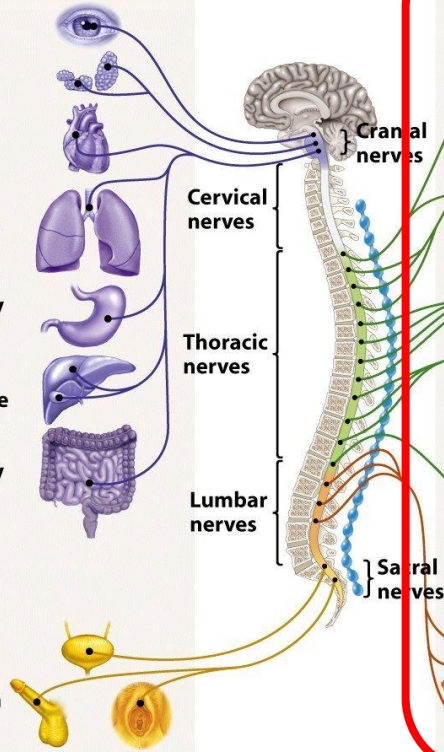
Stimulate activity of stomach

Inhibit release of glucose; stimulate gallbladder

Stimulate activity of intestines

Contract bladder

Promote erection of genitals



SYMPATHETIC NERVES

"Fight or flight"

Dilate pupils

Inhibit salivation

Increase heartbeat

Relax airways

Inhibit activity of stomach

Stimulate release of glucose; inhibit gallbladder

Inhibit activity of intestines

Secrete epinephrine and norepinephrine

Relax bladder

Promote ejaculation and vaginal contraction

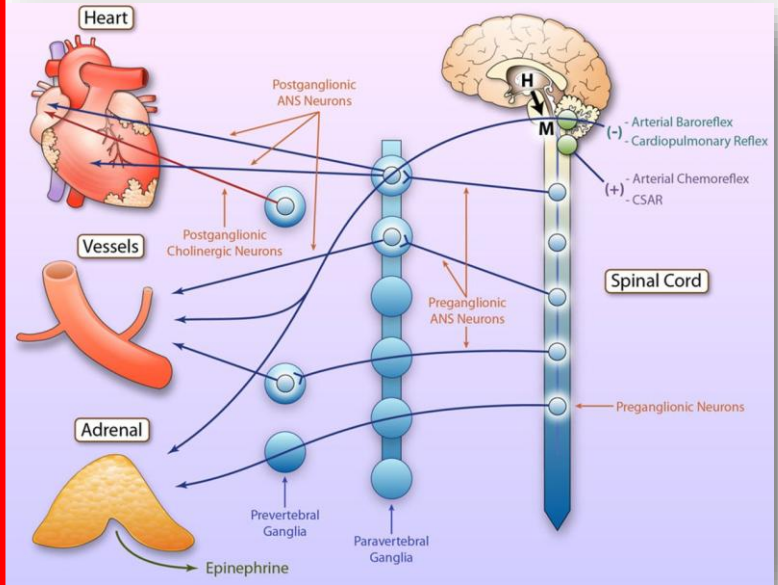
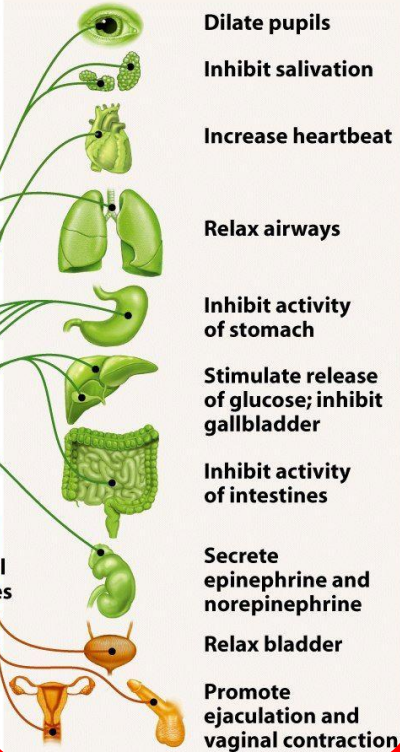


Figure 45-20 Biological Science, 2/e
© 2005 Pearson Prentice Hall, Inc.



B-Adrenoceptors:

1-Alpha receptors

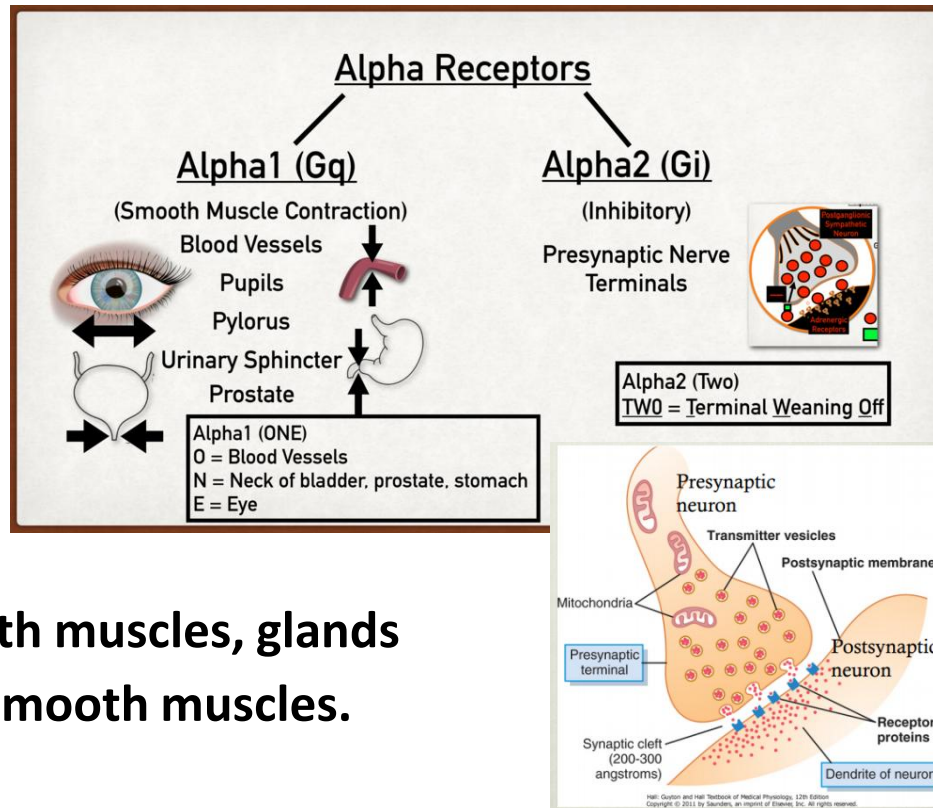
❑ Located in

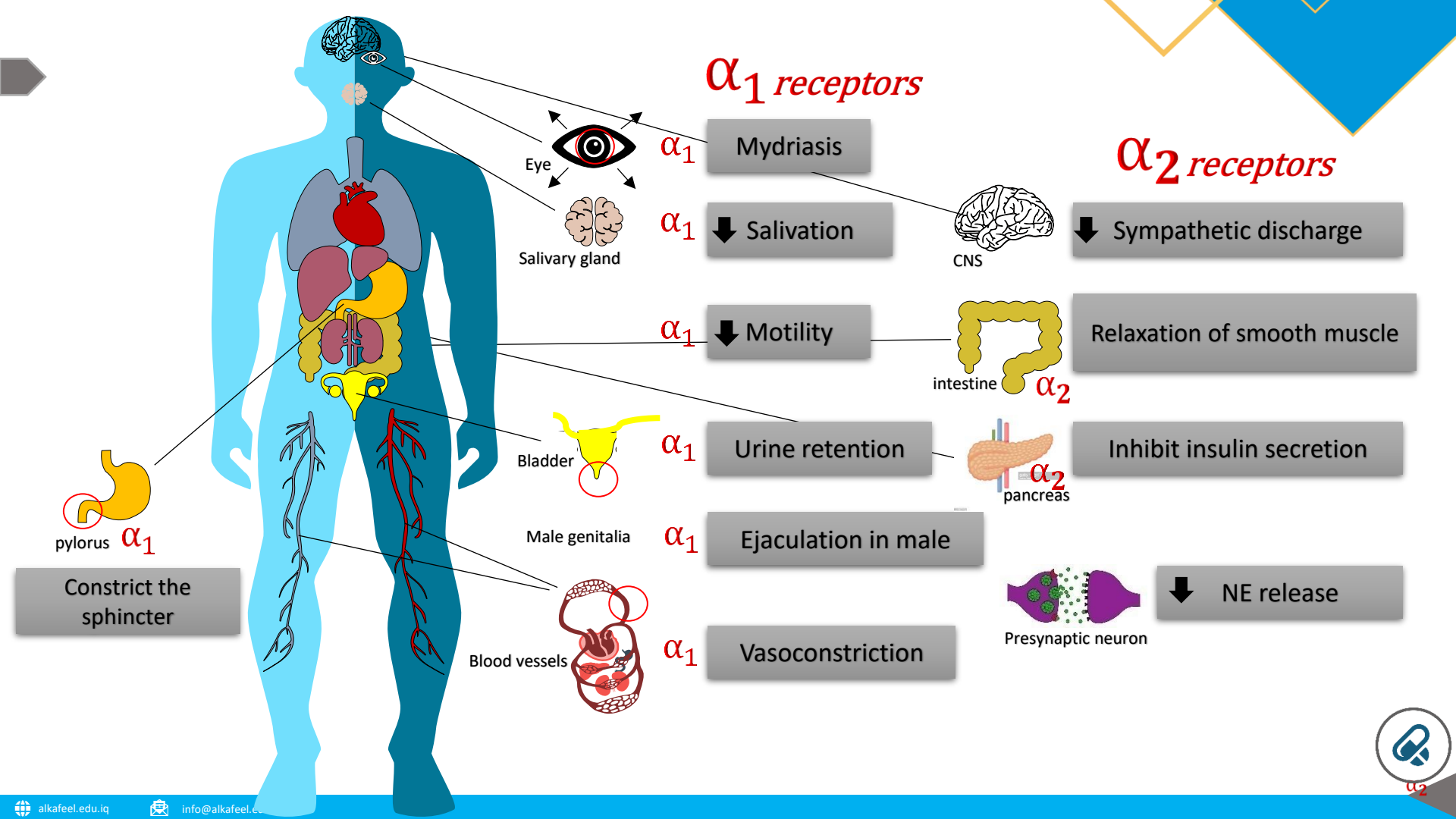
- Blood vessels
- Presynaptic nerve terminals
- Blood platelets
- Fat cells (lipocytes)
- Neurons in the brain

❑ 2 subtypes

➤ **ALPHA1**-effector tissues, smooth muscles, glands

➤ **ALPHA2**-nerve endings, some smooth muscles.





► B-Adrenoceptors:

• 2-Beta receptors

- Located on
 - Most types of smooth muscle
 - Cardiac muscle
 - Some presynaptic nerve terminal
 - Lipocytes
 - Brain
- Subdivided into β_1 β_2 β_3 , they are similar and use the same G-coupling protein
 - ✓ **BETA1**-heart and kidney
 - ✓ **BETA2**-lungs, uterus, liver, heart
 - ✓ **BETA3**-fat or adipose tissue

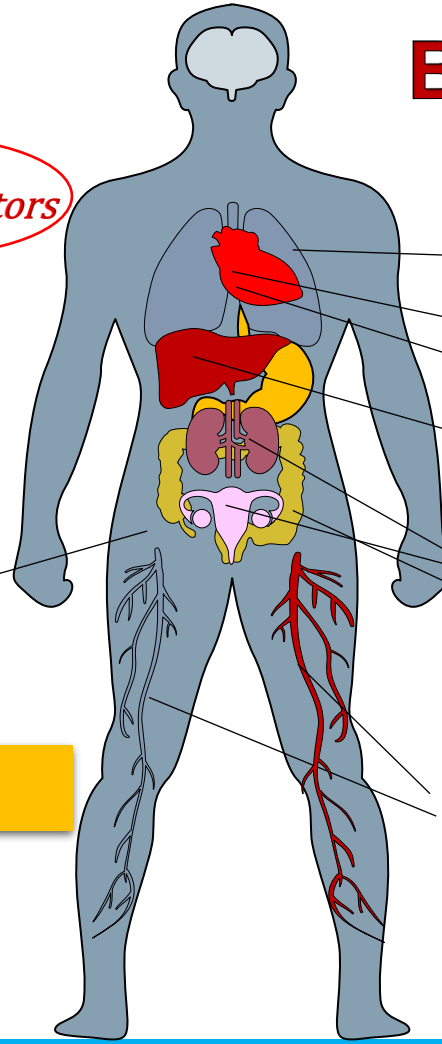


Beta receptors

β_3 receptors

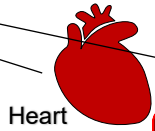
β_1 receptors

β_2 receptors



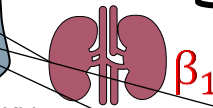
Adipose tissue β_3

↑ Lipolysis



Heart β_1

- ↑ Heart rate
- ↑ atrioventricular contr activity
- ↑ conductivity

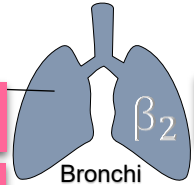


Kidney β_1

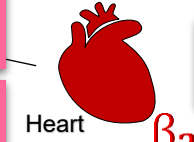
Renin secretion

β_2

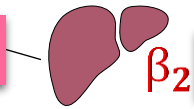
- vasodilation
- arterial dilation



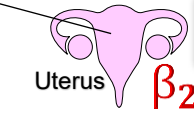
Bronchi β_2



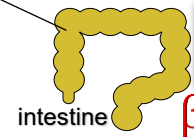
Heart β_2



Liver β_2



Uterus β_2



intestine β_2

Bronchodilation

Relaxation of cardiac muscle

Glycogenolysis

Uterine relaxation

↓ Motility



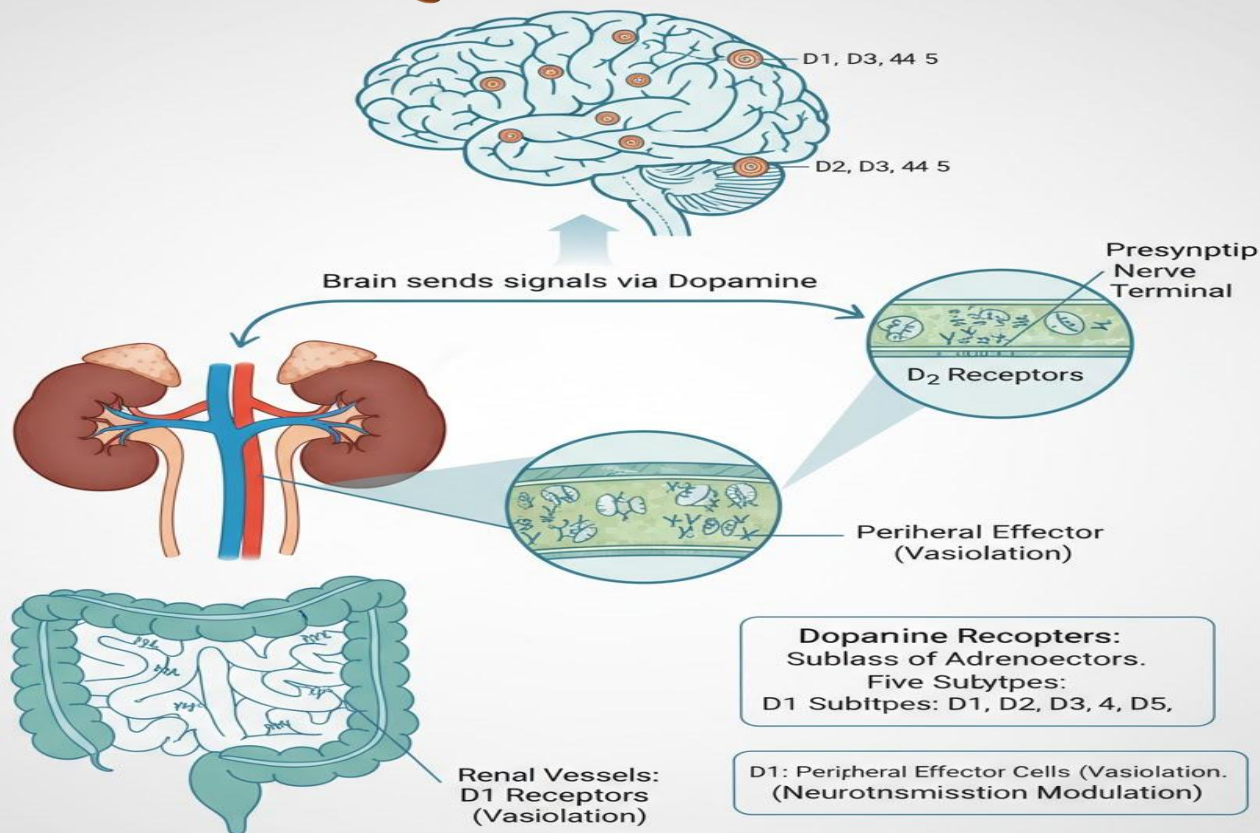
β_2

C-Dopamine receptors

- ❑ Dopamine receptors are subclass of adrenoceptors but they are different in distribution and function
- ❑ They are especially important in renal and splanchnic vessels as well as brain
- ❑ Five subtypes, D1 is the most important dopamine receptor on peripheral effector cells
- ❑ D2 receptors are found in presynaptic nerve terminal



Dopamine receptors





THANK
YOU!

